THE NOVA RESEARCH CURRICULUM

Physics

Grade 8 · Term 3

Student edition

Outcomes: PH-8.3 density, pressure, hydraulics, balanced and unbalanced forces; PH-8.4 stars, galaxies, star life cycles and space exploration.

Lesson notes

- **Density and pressure (Weeks 1–4):** density = mass ÷ volume (g/cm³); water is 1 g/cm³; things less dense than water float. Pressure = force ÷ area, in pascals (Pa = N/m²): sharp knives, snowshoes, camel feet. Liquid pressure rises with depth, so dams are thicker at the bottom.

  ▶ **Try it on NovaSchool:** Density column: float or sink? — open **Simulations** on your dashboard.

- **Hydraulics and forces (Weeks 5–8):** liquids pass pressure on equally, so a small force on a small piston makes a big force on a big piston: car brakes, hydraulic jacks. Balanced forces: no change in motion; unbalanced forces: speeding up, slowing down or changing direction.
- **PH-8.4 Space (Weeks 9–12):** the Sun is a star; galaxies (the Milky Way) hold billions of stars; distances in light-years. Life cycle: nebula → main-sequence star → red giant → white dwarf (stars like the Sun); massive stars → red supergiant → supernova → neutron star or black hole. Space exploration: first Moon landing (1969), the International Space Station, Pakistan's space agency SUPARCO and its satellites.

  ▶ **Try it on NovaSchool:** The life cycle of a star — open **Simulations** on your dashboard.

Workbook pages

Page	Task
1	Density practical: measure mass and volume of five objects.
2	Pressure calculations; dam diagram.
3	Hydraulic jack diagram; balanced vs unbalanced examples.
4	Star life-cycle flow chart.

Question bank

#	Outcome	Question	Your answer
1	PH-8.3	A 200 g block has volume 50 cm³. Its density?	250 g/cm³ / 4 g/cm³ / 0.25 g/cm³ / 150 g/cm³
2	PH-8.3	An object with density 0.8 g/cm³ in water will …	sink / float / dissolve / explode
3	PH-8.3	A 600 N force acts on 2 m². Pressure?	1,200 Pa / 300 Pa / 602 Pa / 3 Pa

#	Outcome	Question	Your answer
4	PH-8.3	Dams are thicker at the bottom because liquid pressure increases with depth. True or false.	True / False
5	PH-8.3	Car brakes use ...	magnets / hydraulics / sound / solar power
6	PH-8.3	If the forces on a moving car are balanced, it ...	stops at once / keeps moving at the same speed / speeds up / turns
7	PH-8.4	Our galaxy is called the ...	Andromeda / Milky Way / Solar System / Orion
8	PH-8.4	A star like the Sun will end as a ...	black hole / white dwarf / comet / planet
9	PH-8.4	A light-year is a unit of ...	time / distance / brightness / mass